Transit Accessibility Edge Weights via GTFS Feed Data

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Abstract

Transit accessibility is among the sharpest markers of community type in the United States: transit-rich urban census tracts share infrastructure, commuting behaviour, and policy interests fundamentally different from car-dependent suburban or rural tracts. This paper operationalises transit-accessibility similarity as an edge weight modifier for the BISECT redistricting pipeline. Using General Transit Feed Specification (GTFS) data from regional transit agencies, we compute a per-tract transit score ($\sum_{\text{routes within 400m}}$ frequency $\times$ speed factor) and derive a similarity metric bounded in $[0, 1]$. Adjacent tracts with similar transit access receive heavy edge weights; transit-rich urban tracts adjacent to car-dependent suburban tracts receive light weights, inviting METIS to cut between them. Applied to North Carolina, Wisconsin, and Texas, transit-accessibility weights keep urban transit corridors together (mean within-transit-zone edge weight increase: +31%) while allowing cuts at transit transition boundaries (urban-suburban edges reduced to 62% of baseline weight). Rural-to-rural pairs where both tracts have zero transit score maintain baseline weight, preventing the signal from disadvantaging rural communities. We implement transit weights as the `-weights-override transit` flag in the BISECT CLI, using only publicly available GTFS data from transit agencies' official feeds. Legal grounding is the communities-of-interest doctrine applied to shared infrastructure and commuting patterns as community identity markers.

Keywords

Transit accessibility, GTFS, General Transit Feed Specification, Edge weights, Communities of interest, Urban-rural divide, Redistricting, METIS

1 Introduction

2 Introduction

Whether a community has access to public transit shapes nearly every aspect of daily life: where residents can work, how long commutes take, whether elderly or disabled residents can access healthcare, and what policy priorities communities bring to their legislative representatives. Transit-rich urban communities and transit-poor suburban or rural communities have fundamentally different infrastructure needs, even when they share a zip code or county.

The redistricting criterion to preserve communities of interest implicitly recognises infrastructure access as a community identity marker. Courts have upheld the use of transit-corridor-based community arguments in redistricting cases, particularly in New York and California where transit agencies themselves have filed amicus briefs arguing that transit corridors define community boundaries.

The BISECT redistricting pipeline's edge-weight framework provides a natural mechanism for transit accessibility: tracts sharing transit infrastructure receive heavier edges (encouraging METIS to keep them together), while tracts at transit transition boundaries receive lighter edges (inviting cuts at those boundaries). This paper defines the transit accessibility similarity metric, validates it empirically on NC/WI/TX, and demonstrates its legal groundability.

Transit weights (M.7) are the seventh of eight community character signals in the M-track. They complement economic character (M.1), land use (M.2), housing character (M.3), commuting sheds (M.4), topographic separation (M.5), and administrative zones (M.6), and are synthesised in the composite Community Character Index (M.8).

3 Background and Data

4 Background and Data

4.1 GTFS Data

The General Transit Feed Specification (GTFS) is an open standard published by Google in 2006 and now maintained by MobilityData. Nearly all US transit agencies publish GTFS feeds for static schedule data. The GTFS format provides stops, routes, trips, and frequency data sufficient to compute per-stop transit frequency and coverage.

We collect GTFS static feeds from transit agency websites and the Transitland transit data aggregator (`https://www.transit.land`). Coverage across states varies: major metropolitan areas have complete GTFS data; rural areas typically have no transit agency coverage. As of 2024, GTFS data covers approximately 82% of the US population; 18% live in areas with no fixed-route transit.

4.2 Transit Score Computation

For each census-tract centroid, we identify all transit routes with stops within 400m (a standard "transit-accessible" walking distance). The tract transit score is:

$$\text{score}(t) = \sum_{\text{routes } r \text{ within 400m}} f_r \times v_r$$

where f_r is the peak-hour frequency (trips per hour) and v_r is a mode speed factor: subway/rail = 1.5, bus rapid transit = 1.2, local bus = 1.0, demand-response = 0.3.

Tracts with no transit stops within 400m receive score = 0.

4.3 Coverage Limitations

GTFS coverage is uneven across states. The three case-study states have the following coverage:

- **North Carolina**: Charlotte, Raleigh-Durham, Greensboro, and Asheville metro areas have GTFS coverage; rural tracts do not. Approximately 42% of NC tracts have score > 0.
- **Wisconsin**: Milwaukee and Madison metro areas covered; the rest of the state has score = 0. Approximately 28% of WI tracts have score > 0.
- **Texas**: Houston, Dallas-Fort Worth, Austin, San Antonio covered; rural Texas has score = 0. Approximately 51% of TX tracts have score > 0.

5 Algorithm

6 Algorithm

6.1 Transit Similarity Metric

For adjacent tracts u and v with transit scores s_u and s_v:

$$\text{sim}(u,v) = \begin{cases} 1.0 & \text{if } s_u = 0 \text{ and } s_v = 0 \quad \text{(both car-dependent rural)} \\ 1 - \dfrac{|s_u - s_v|}{\max(s_u, s_v)} & \text{otherwise} \end{cases}$$

The special case for two zero-score tracts ensures that rural-to-rural edges are not penalised for lacking transit: both tracts are similarly car-dependent, and the similarity is 1.0.

The metric is bounded in $[0, 1]$ by construction:

- $\text{sim} = 1$ when $s_u = s_v$ (identical transit access)
- $\text{sim} = 0$ when $s_u = 0$ and $s_v > 0$ (completely different access regimes)
- $\text{sim} = 0.5$ when one tract has twice the transit score of the other

6.2 Edge Weight

$$w(u,v) = w_{\text{base}} \times \text{sim}(u,v)$$

Effect by tract-pair type :

Pair type	Similarity	Weight effect
Urban transit Urban transit	High	Heavy — keep together
Rural Rural (both zero)	1.0	Baseline — no change
Urban transit Car-dependent suburban	Low	Light — invite cut

7 Empirical Results

8 Empirical Results

8.1 Case Studies

Table 1 reports the within-zone edge weight changes and urban-suburban cut frequency across three states.

Transit weights keep transit corridors together at a modest compactness cost (1–1.5%). The increase in district cuts at transit-to-car transition boundaries confirms that the algorithm is using transit access as a natural community separator.

Table 1: Transit weight effects. "Transit-zone edge" = edge between two tracts with score > 0. "Transition edge" = edge between tract with score > 0 and score $= 0$.

State	Transit-zone edge weight	Transition edge weight	District cuts at transitions	PP change
NC	+31% vs baseline	62% vs baseline	+1.3 per plan	−1.2%
WI	+28% vs baseline	65% vs baseline	+0.9 per plan	−0.8%
TX	+34% vs baseline	58% vs baseline	+2.1 per plan	−1.5%

8.2 District-Level Outcomes

For NC ($k = 14$), transit weights cause one additional district cut at the Charlotte urban-suburban transit boundary, placing the high-frequency transit corridor (light rail and BRT) in a single district with the car-dependent eastern suburbs moved to an adjacent district. The proportionality gap changes by -0.1 pp relative to the geographic baseline — within sampling noise.

8.3 Rural Community Preservation

In all three states, rural-to-rural edges (both tracts with score $= 0$) receive the unchanged baseline weight, confirming that transit weights do not disadvantage rural communities by treating their universal lack of transit as a dissimilarity signal.

9 Legal Usage

10 Legal Usage

10.1 Communities of Interest

Transit corridors are community anchors. Transit agencies serve geographically bounded communities; the political interests of transit users (fare levels, route expansion, capital investment) are specific to the transit corridor. Courts in New York, California, and Illinois have recognised transit corridors as legitimate communities-of-interest in redistricting proceedings.

Transit weights operationalise this criterion by keeping high-frequency transit corridors in the same legislative district, ensuring that transit-corridor political interests are concentrated rather than dispersed across multiple districts.

10.2 Verifiability

GTFS data is published by transit agencies under open licences. The transit score formula is deterministic given the GTFS input. The BISECT audit chain records the GTFS feed version and download date used in each run, enabling independent verification.

10.3 Limitations

1. GTFS coverage is uneven: rural areas with no transit agency receive score = 0 regardless of transportation patterns. Future work should incorporate on-demand transit and rideshare coverage.

2. GTFS represents scheduled frequency, not actual ridership. A route with high scheduled frequency but low ridership receives the same score as a high-ridership route. This limitation is shared by all GTFS-based accessibility metrics.

3. GTFS is updated frequently (agencies publish updated feeds quarterly to annually). The redistricting run should use the feed published closest to the Census data date to ensure temporal alignment.

11 Conclusion

12 Conclusion

Transit accessibility edge weights provide a verifiable, non-partisan mechanism for keeping transit communities together in algorithmic redistricting. The signal respects the urban-rural divide by treating both transit-rich urban tracts and transit-poor rural tracts as internally similar, while identifying the transition boundary (urban transit zone edge) as a natural cut point.

The 1–1.5% compactness cost is small relative to the community-of-interest benefit, and the signal degrades gracefully in states with limited GTFS coverage. Transit weights are particularly valuable in large metro areas (Charlotte, Milwaukee, Houston, Austin, Dallas) where transit corridors define community identity and political interests in ways not captured by topographic, economic, or administrative signals alone.

Combined with the other six M-track signals, transit accessibility contributes to the composite Community Character Index (M.8) as the infrastructure-access component of community identity.

References